



Handout Garmin G1000 Part 1 Area Navigation Basics and „PBN“

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Some images used in this program are generic pictures and will not necessarily represent the current state of hardware, software or approach procedures.

Garmin 1000 Part 1 Area Navigation Basics and

„PBN“: Performance Based Navigation

Credit:
Parts of this teaching material are taken from
- Garmin
- UK CAA document CAP 733
- ICAO PBN Manual
- FSF Seminar 2011
- ICAO PBN Workshop Tanzania
- Wikipedia



Vers. 11

What is “area navigation” ?

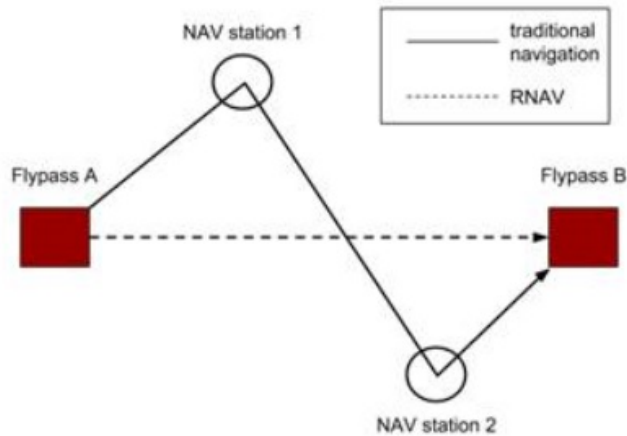
■ Area Navigation

- a method of navigation that permits aircraft operation **on any desired course**

but:

- within the coverage of station-referenced navigation signals or within the limits of a self contained system, or a combination of these.

What is “area navigation” ?



„RNAV“

- For long time the acronym „RNAV“ was used as an abbreviation for „area navigation“ in a general sense
 - But observe:
- Now „RNAV“ is a specific designation for one of the navigation functionalities within the new „PBN“ concept

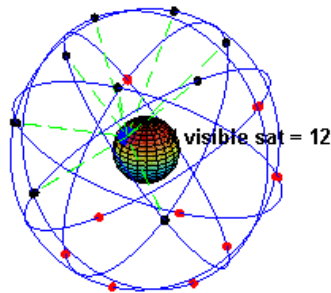
Global Navigation Satellite Systems (GNSS)

GNSS Space segment

- **GNSS Systems:**
 - US American NAVSTAR **GPS**
 - Russian **GLONASS**
 - European GALILEO is not yet operational

GPS space segment

- The satellites are in the „WGS 84 system“
- 24 operable satellites in six orbits
- Orbit height 20.200 km
- The „satellite almanach“ contains the orbital data

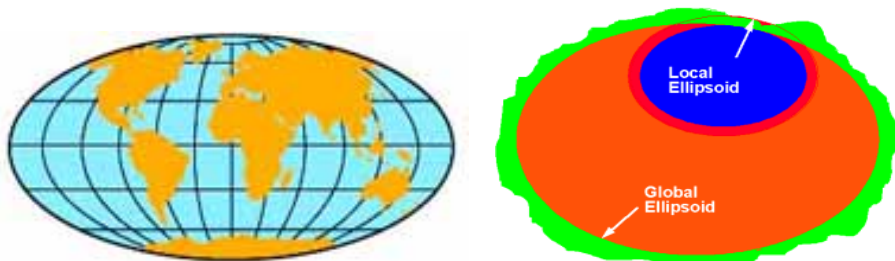


GLONASS space segment

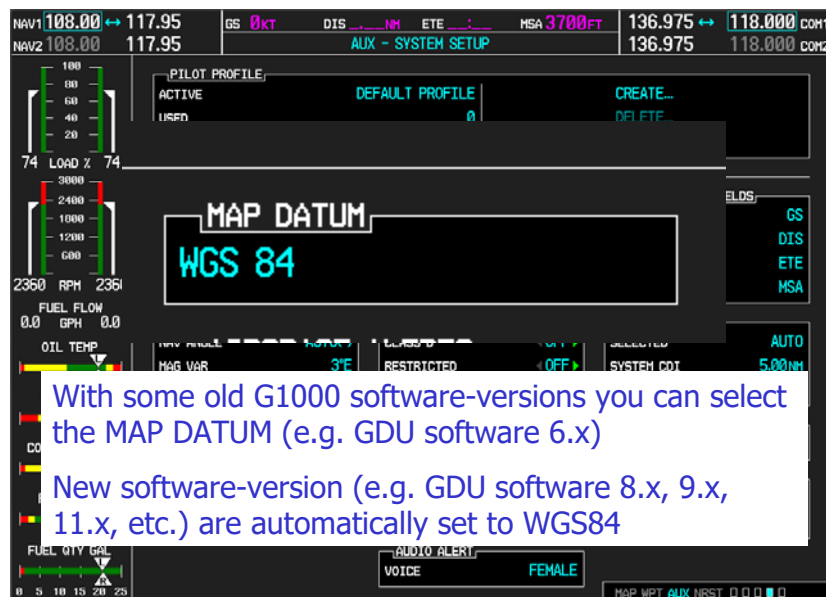
- 24 operable satellites in three orbits
- Orbit height 19.100 km
- The satellites are in the „PZ-90 (Parametri Zemli 90) system“. System parameters differ only slightly from WGS 84

- Mathematically simplified the earth is an „Ellipsoid of rotation“
- By projecting this curved surface to a flat topographical map, distortions are created

- Distortions are minimized by using local parameters for the ellipsoid



- Therefore: many different reference systems („map datum“)
- A specific point will have different „coordinates“ in different reference systems
- GPS is based on the reference system **WGS 84**
 - Therefore:
- All coordinates must be published according to „WGS 84“





WARNING: Do not use GPS to navigate to any active waypoint identified as a 'NON WGS84 WPT' by a system message. 'NON WGS84 WPT' waypoints are derived from an unknown map reference datum that may be incompatible with the map reference datum used by GPS (known as WGS84) and may be positioned in error as displayed.

NON WGS84 WPT – Do not use GPS for navigation to [xxxx]

The position of the selected waypoint [xxxx] is not calculated based on the WGS84 map reference datum and may be positioned in error as displayed. Do not use GPS to navigate to the selected non-WGS84 waypoint.

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GPS Altitude

- In reality the earth is neither a perfect sphere, nor the mathematically simplified „Ellipsoid of rotation“,



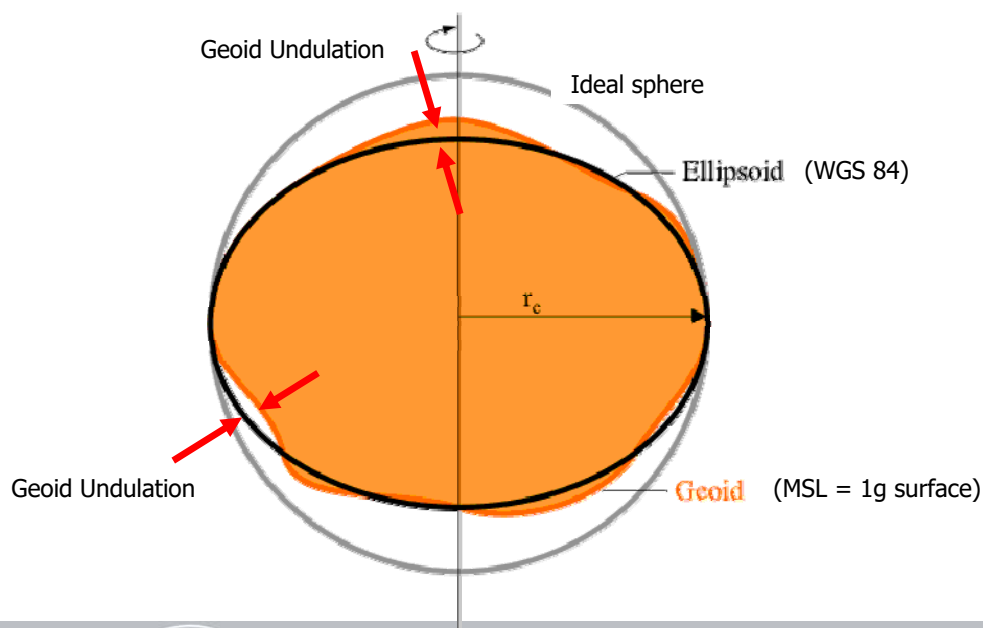
- but the earth's surface is the „Geoid“

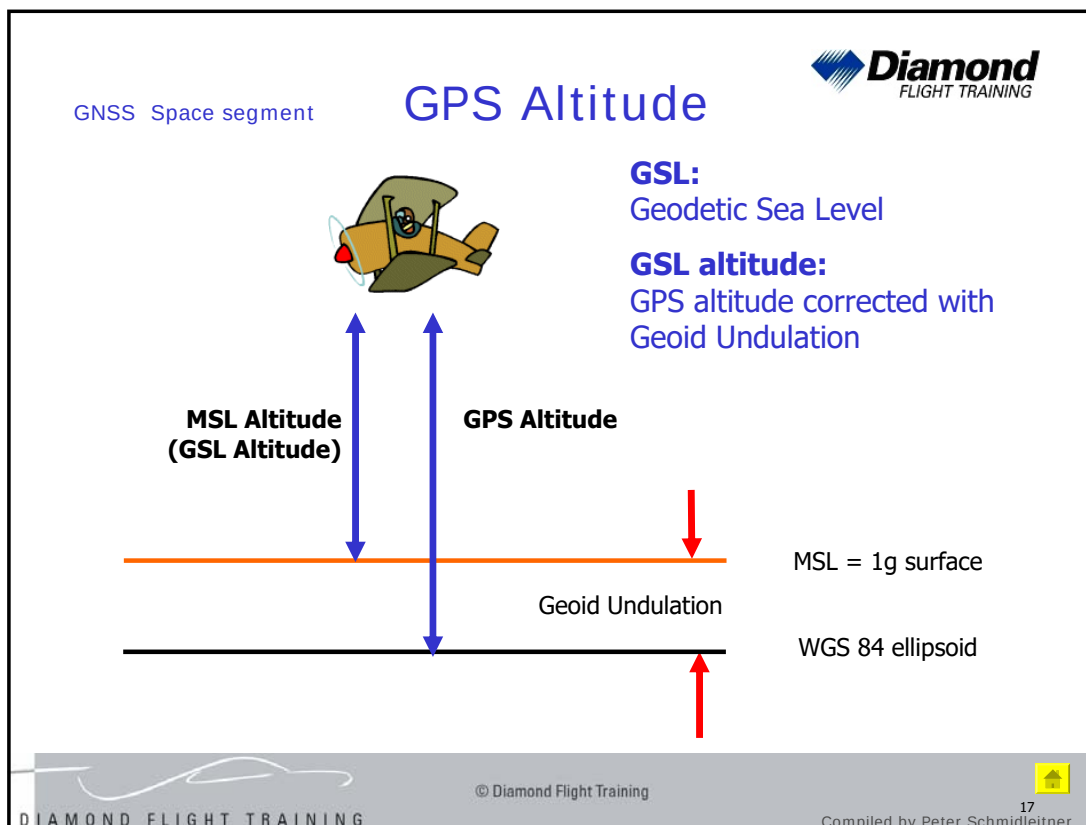
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- With BARO altimetry we measure the altitude above the „Geoid“ (MSL) *)
- GPS expresses the altitude above the „WGS 84“ ellipsoid
 - *) what is MSL?
 - MSL is the 1g surface





GNSS Space segment

GPS Altitude

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LUFTFAHRTHANDBUCH ÖSTERREICH
AIP AUSTRIA

LOWW 2-1
27 AUG 2010

LOWW AD 2.1 ORTSKENNUNG UND NAME DES FLUGPLATZES
LOWW AD 2.1 AERODROME LOCATION INDICATOR AND NAME

LOWW - WIEN SCHWECHAT

LOWW AD 2.2 LAGE UND VERWALTUNG DES FLUGPLATZES
LOWW AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	KOORDINATEN UND LAGE DES FLUGPLATZBEZUGSPUNKTES ARP COORDINATES AND SITE AT AD	N 48 06 37 E 016 34 11 1483 M 157° GEO vom Kontrollturm N 48 06 37 E 016 34 11 1483 M 157° from control tower
2	RICHTUNG UND ENTFERNUNG VON WIEN DIRECTION AND DISTANCE FROM WIEN	9 NM südöstlich von Wien 9 NM southeast from Wien
3	FLUGPLATZHÖHE ÜBER MEERESSPIGEL/BEZUGSTEMPERATUR ELEVATION/REFERENCE AD TEMPERATURE	183 M (600 FT) / 25.7° C
4	GEOID UNDULATION	44 M / 144 FT
5	ORTSMESSWEISUNG/ JÄHRLICHE ÄNDERUNG MAG VARIATION/ANNUAL CHANGE	3° E / 2010

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GPS Status Page



Terrain Page



GPS Control segment

- 10 Monitoring Stations
 - Monitoring Stations transmit their data to the Master Control station.
- Master Control Station
 - The Master control Station is responsible for satellite constellation surveillance and control. It is located in Colorado.

GLONASS Control segment

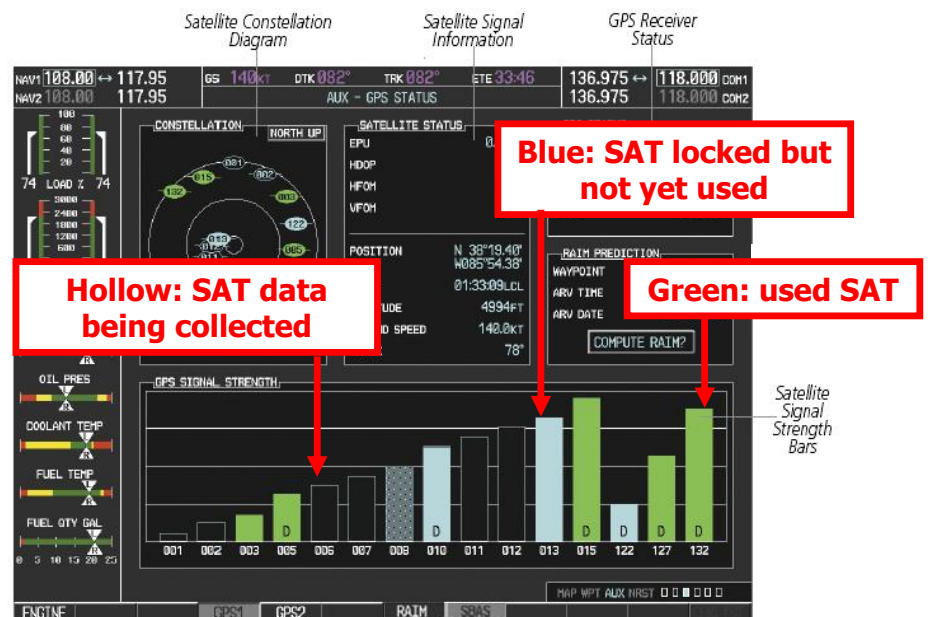
- 5 Monitoring Stations
 - Monitoring Stations transmit their data to the Master Control station
- Master Control Station in Moscow
 - The Master control Station is responsible for satellite constellation surveillance and control

GPS User Segment

- GPS elements
 - Antenna
 - Receiver
 - multi channel; tracks the orbits of all satellites that are visible to it
 - receiver position is stored on shut-down; last satellite constellation can be used on start-up within 4 hours, thereafter the receiver will scan for receivable satellites and get new almanach data

GPS User Segment

- GPS supplies
 - position fixes
 - 3 satellites 2D
 - 4 satellites 3D
 - speed data
 - time reference



GNSS Augmentation Systems

- SBAS
- GBAS

- What is the difference between GBAS (DGPS, Differential GPS) and SBAS?:
 - GBAS (DGPS): the correction data measured at the reference-receiver are not necessarily valid for the present position
 - SBAS: calculates a „ionosphere correction grid“ from data of various RIMS

- **SBAS:**
Satellite Based Augmentation System
 - **WAAS**
 - Wide Area Augmentation System (USA)
 - **EGNOS**
 - European Geostationary Navigation Overlay System
 - **MSAS**
 - Multi-functional Satellite Augmentation System (Japan)
 - **GAGAN**
 - GPS and GEO Augmented Navigation (India)
 - **SDCM**
 - System vor Differential Corrections and Monitoring (Russia)



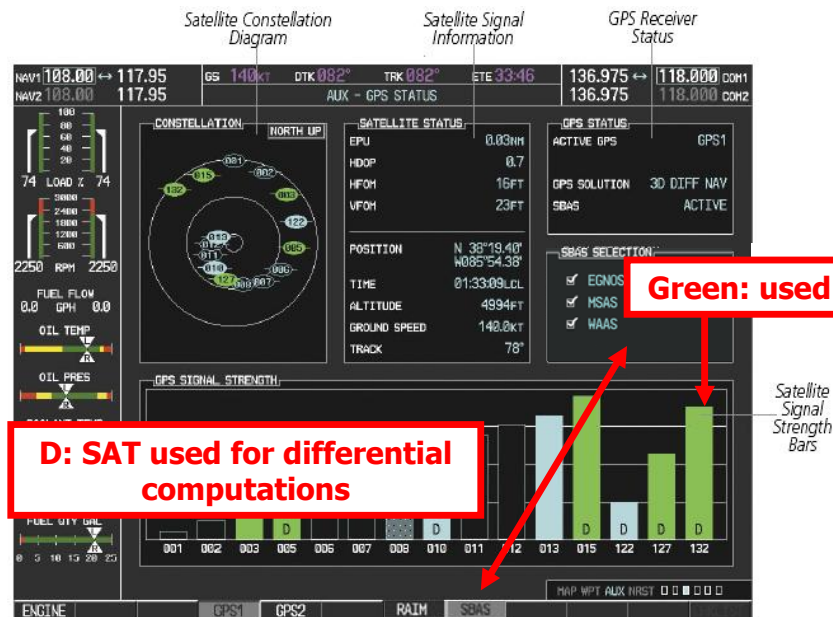


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- Very often „WAAS“ is used as a generic reference to SBAS systems.
- SBAS services are „interoperable“; receivers interpret whichever signal(s) they „see“.

- RIMS: Ranging and Integrity Monitoring Stations
 - WAAS: 37 Stations
 - EGNOS: 39 Stations
- SBAS correction signals are broadcast via geostationary satellites (EGNOS 3 satellites)

- For aviation EGNOS provides:
„SoL“ service („Safety of Life“ service)
based on the integrity data
- 2 service levels:

	Enroute + NPA	APV
Lateral accuracy	220 m	16 m
Vertical accuracy	n.a.	20 m
Horizontal alert limit	56 m	40 m
Vertical alert limit	n.a.	50 m
Availability	99%	99%

PBN Principles

The PBN concept:

A shift from **sensor-based** to
performance-based navigation

Area navigation is based on performance requirements expressed in

Navigation Specifications

in terms of

- Accuracy
- Availability
- Continuity
- Functionality
- Integrity

■ **Accuracy**

The degree of conformance between

- estimated, measured, or desired position and/or velocity
 - and
- true position or velocity.

Accuracy indications on G1000

Satellite Status	
EPU	0.03NM
HDOP	0.7
HFOM	0FT
VFOM	23FT

- EPU
 - Estimated Position Uncertainty
 - Radius of circle around estimated position with 95% probability. Statistical value, not actual error measurement
- DOP
 - Dilution of Precision
 - Satellite geometry quality.
0.0 good → 9.9 bad
- HFOM, VFOM
 - Horizontal, Vertical figures of merit.
 - Position uncertainty; 95% confidence values reported by the GPS receiver

■ Availability

An indication of the ability of the system

- to provide usable service within the specified coverage area
- defined as the portion of time during which the system is to be used for navigation.

■ Continuity

The capability of the total system to perform its function without non-scheduled interruptions during the intended operation.

- Loss of function is classified as a minor failure condition if the operator can revert to a different navigation system and safely proceed to a suitable airport.

■ Functionality

The ability of a system to perform a task or function.

■ Integrity

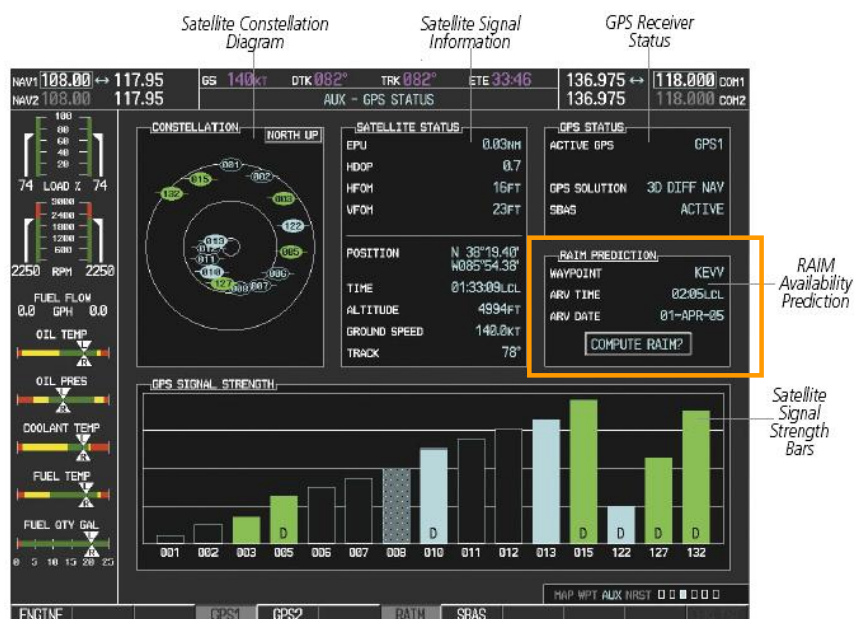
The ability of a system to provide timely warnings to users when the system should not be used for navigation.

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R eceiver A utonomous I ntegrity M onitoring

- **Receiver Autonomous Integrity Monitoring (RAIM)**
- A technique whereby a GNSS receiver/processor determines the integrity of the GNSS navigation signals using only GPS signals or GPS signals augmented with altitude.
- At least one satellite in addition to those required for navigation should be in view for the receiver to perform the RAIM function.

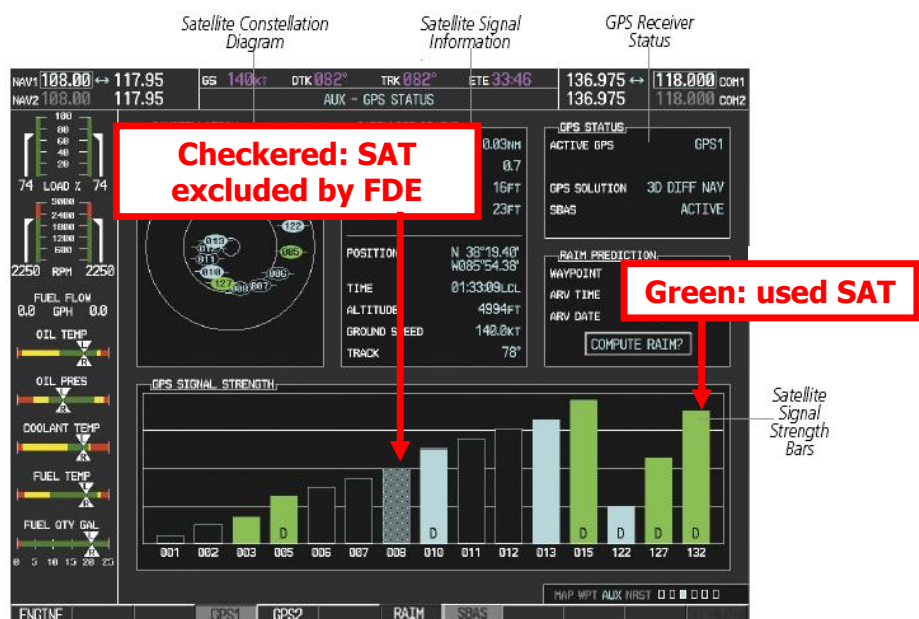
- 5 SATs necessary for RAIM
- SAT geometry must allow accuracy of
 - 2 NM oceanic and en-route
 - 1 NM terminal
 - 0,3 NM approach



F ault D etection and E xclusion

- Fault Detection:
detects presence of unacceptable large error
- Fault Exclusion:
excludes source of unacceptable large error

- 6 SATs necessary to eliminate corrupt SAT



- Pre departure verification of FDE:
 - Necessary for oceanic/remote operation with GPS as „sole means of navigation“
 - G1000: Combined with RAIM prediction

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PBN Components

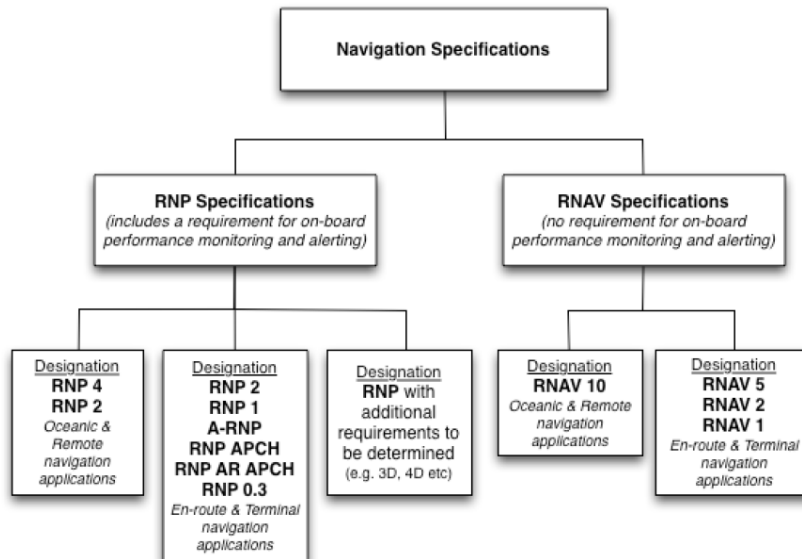
PBN components

- The PBN Components are:
 - NAVAID infrastructure
 - Navigation Specification
 - Navigation application

■ Navaid infrastructure:

- GPS is the primary navigation system to support RNP APCH Procedures.
- The missed approach segment may be based on the conventional Navaid (e.g., VOR, DME, NDB).

- The **Navigation Specification** is either an
 - **RNP** specification
 - or
 - **RNAV** specification



- Precision Approaches and landing systems like ILS, MLS are not included in the concept of PBN, because they are not based on area navigation techniques.

■ Navigation application:

- **Oceanic, en-route, terminal**
 - Linear performance requirements
- **Approach phase**
 - Both linear and angular laterally guided operations

NAVIGATION SPECIFICATION	FLIGHT PHASE							DEP
	En Route OCEANIC /REMOTE	En Route Continental	ARR	APPROACH				
				Initial	Interm.	Final	MISSED	
RNAV 10	10							
RNAV 5		5	5					
RNAV 2		2	2					2
RNAV 1		1	1	1	1		1 ^b	1
RNP 4	4							
Basic-RNP 1			1 ^{a,c}	1 ^a	1 ^a		1 ^{ab}	1 ^{a,c}
RNP APCH				1	1	0.3	1	
RNP AR APCH				1-0.1	1-0.1	0.3 – 0.1	1-0.1	

Difference between RNAV and RNP

RNP – RNAV Difference

- **RNP** specification includes a requirement for on-board self-contained monitoring and alerting
- **RNAV** relies on external monitoring, adequate route spacing and ATC intervention

Functional requirements for RNAV and RNP

Functional requirements

- Both RNAV and RNP must provide:
 - Continuous indication of
 - Lateral deviation
 - Distance/bearing to active waypoint
 - Groundspeed or time to active waypoint
 - Navigation data storage (database)
 - Failure indication

Designation of RNAV and RNP Specifications

Designation

- Designation of RNAV or RNP:
 - RNAV x
 - or
 - RNP x
 - where „x“ is the lateral navigation accuracy in NM, i.e. the **Total System Error**, expected to be achieved at least 95% of flight time

Use of PBN

Use of PBN

- The concept of PBN is used for:
 - Airspace planning
 - Minimum route spacing
 - Airworthiness approval
 - Operational approval

The concept of „Path Terminator“

Path Terminator

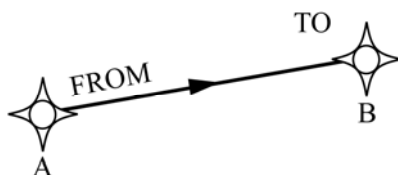
- The so called „Path Terminators“ are the „leg types“, from which procedures are compiled.
- Observe!
 - The mentioning of a conventional NAV aid in the Path Terminator description does not mean that this NAV aid must be operational!

- Most important for PBN are
 - IF, TF, CF, DF, FA, CA
 - RF and FRT
 - The concept of „fly by“ and „fly over“
 - Offset flight path

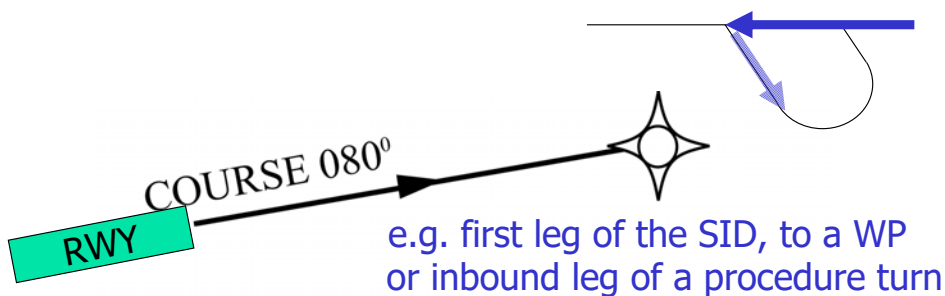
Initial Fix (IF)



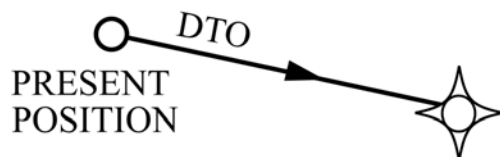
Track to Fix (TF) (between two fixes)



Course to Fix (CF)



Direct to Fix (DF)



Fix to Altitude (FA)



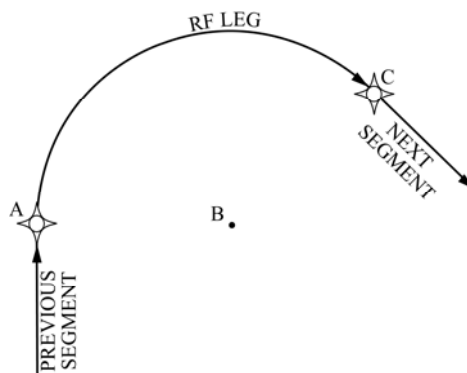
e.g. first leg of the SID, to an altitude

Course to Altitude (CA)

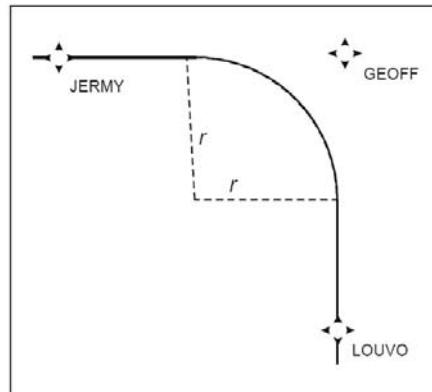


e.g. first leg of the SID, to an altitude

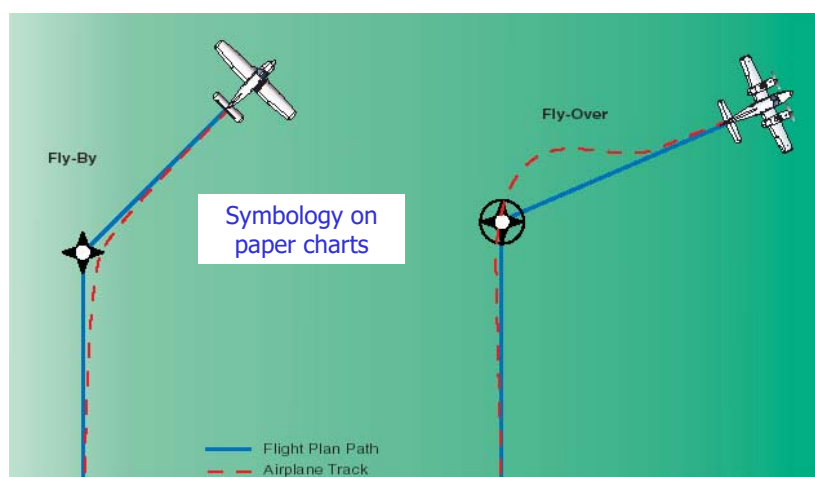
Radius to a Fix (RF)



Fixed Radius Transition (FRT)



Fly-by / Fly-Over



G1000:

No indication whether a waypoint is „fly by“ or „fly over“

- by the waypoint symbol on the maps
- in the flight plan

Only indicated by the

provided the map
range is appropriate
to make it visible

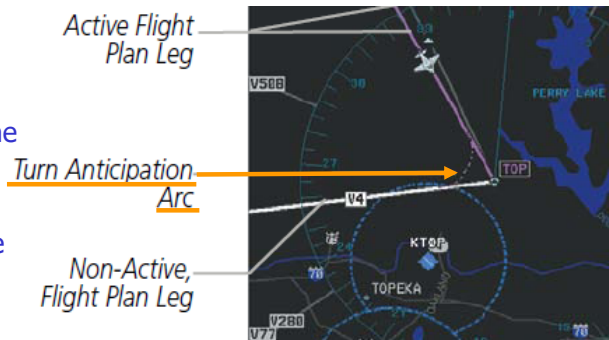
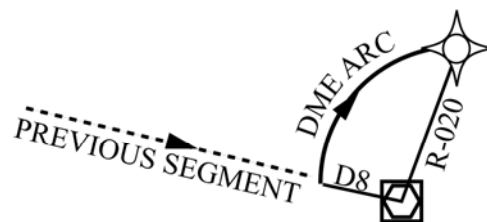
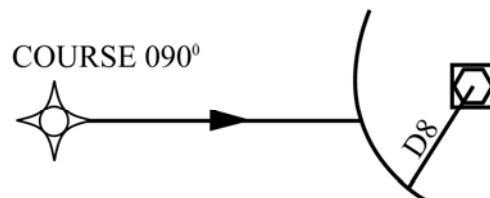


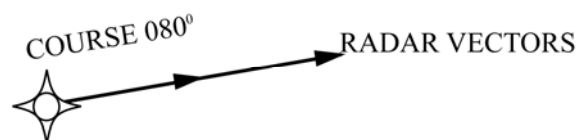
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Arc to a Fix (AF)

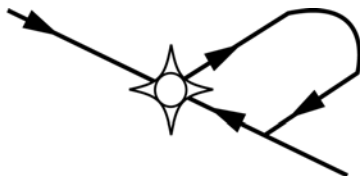
Fix to DME termination (FD)



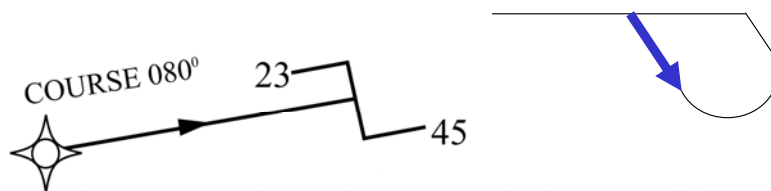
Fix to Manual termination (FM)



Procedure turn to Intercept (PI)



Fix to distance on Course (FC)

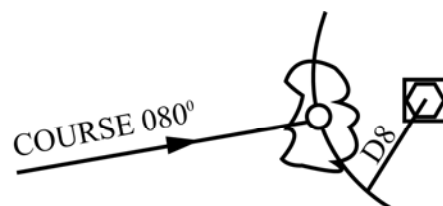


e.g. outbound leg of a procedure turn

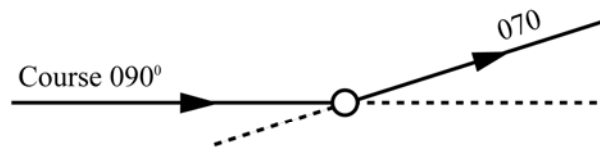
***Hold to Fix (HF), Hold to Altitude (HA),
Hold to Manual termination (HM)***



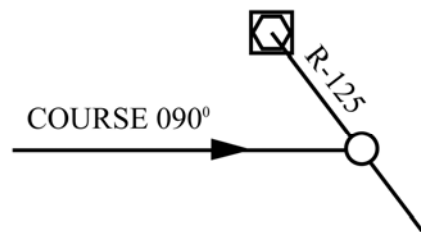
Course to DME termination (CD)



Course to Intercept (CI)



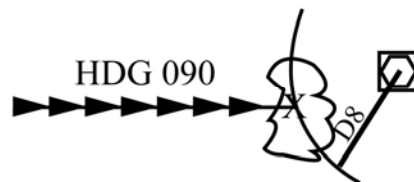
Course to Radial interception (CR)



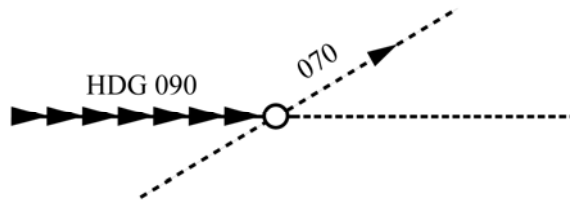
Heading to Altitude (VA)



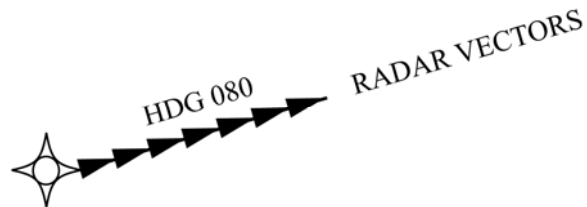
Heading to DME distance (VD)



Heading to next leg Intercept (VI)



Heading to Manual termination (VM)



Heading to Radial termination (VR)



- Heading legs:
 - Not all databases / NAV programs support “heading legs”
 - Most RNAV-equipped aircraft fly tracks. Take care if “heading” is required for ATC separation

G1000:

If HDG legs are supported:

FLIGHT PLAN		
LOWG / XANUT		
	DTK	DIS
RW35C		
2000FT	___°	___NM
→ INTRCPT	hdg 089°	16.9NM
XANUT	060°	10.0NM



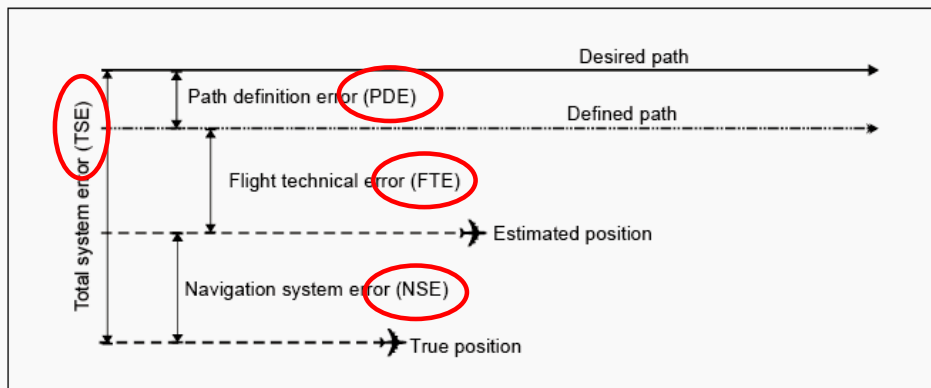
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Assumed errors in the PBN concept

Errors

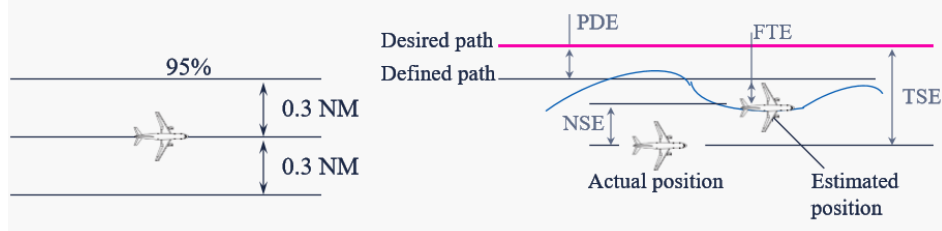
- Path definition error
- Flight technical error
- Navigation system error
- Total system error

■ Lateral navigation errors (95%)

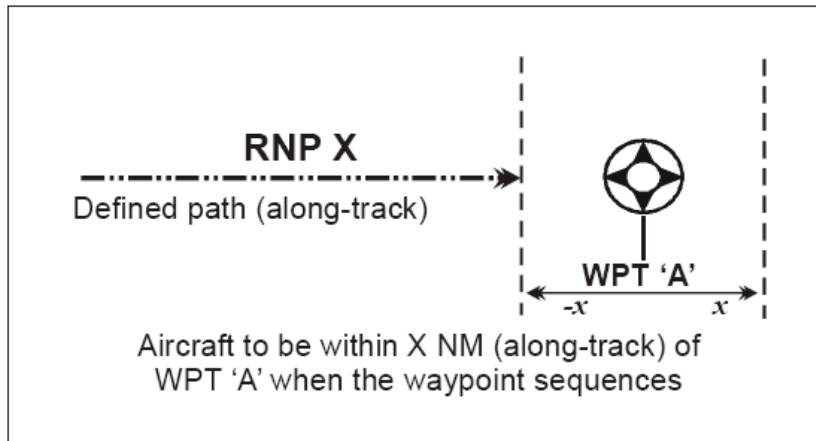


Note: NSE is sometimes referred to as positioning estimation error (PEE).

- The lateral TSE must be within ± 1 NM (95%) for the initial, intermediate and missed approach segment
- The lateral TSE must be within ± 0.3 NM (95%) for the Final segment
- The Total System Error is the Root Sum Square (RSS) of:
 - Navigation system error (NSE)
 - Path definition error (PDE)
 - Flight technical error (FTE)

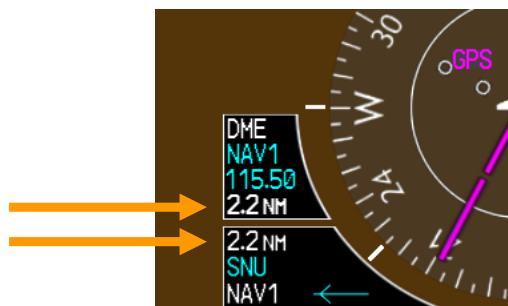


■ Along-track navigation errors (95%)



■ How to detect an inaccuracy without on-board monitoring?

- Compare RNAV data with „raw data“
 - best method: compare DME distances
 - e.g.:
 - if you have a VOR/DME location as a WPT,
 - select same VOR/DME as „NAV“-source,
 - compare „DME reading“ with „Dist to WPT“



**A so called „map shift“
is very unlikely with
SBAS**

Abnormal situations

Abnormal situations

- For RNP on-board performance monitoring and alerting is required
- In case of loss of PBN capability abnormal and contingency procedures have to be used
- Where the contingency procedure requires reversion to conventional navigation, necessary preparations must be completed before commencing the RNP procedure
- If a Missed Approach has to be initiated because GPS guidance is lost, and no conventional NAV aids are available, follow DR (Dead Reckoning)

- The pilot must notify ATC when the RNAV performance ceases to meet the requirements
- The communication to ATC must be in accordance with the authorized procedures
- In the event of communications failure, the pilot should continue with the published com-fail procedure

Samples of on-board alerts

GPS NAV LOST – Loss of GPS navigation. Insufficient satellites.	Loss of GPS navigation due to insufficient satellites.
GPS NAV LOST – Loss of GPS navigation. Position error.	Loss of GPS navigation due to position error.
GPS NAV LOST – Loss of GPS navigation. GPS fail.	Loss of GPS navigation due to GPS failure.
LOI – GPS integrity lost. Crosscheck with other NAVS.	GPS integrity is insufficient for the current phase of flight.

Samples of on-board alerts

Warning alerts (red)	Meaning / Cause
WARN	RAIM position warning. The nav deviation bar is removed.

Caution alerts (amber)	Meaning / Cause
LOI	GPS integrity is insufficient for the current phase of flight.

Samples of on-board alerts

Warning Alerts	Meaning / Cause
POSN ERROR	The annunciation is active when the G1000 will no longer provide GPS based navigational guidance.

Caution Alerts	Meaning / Cause
INTEG RAIM not available	The annunciation is active when RAIM (Receiver Autonomous Integrity Monitor) is not available.

Samples of on-board alerts

Annunciation	Location	Description
GPS LOI	Right of HSI	Loss of Integrity Monitoring–GPS integrity is insufficient for the current phase of flight
GPS INTEG OK	Right of HSI	Integrity OK–GPS integrity has been restored to within normal limits (annunciation displayed for 5 seconds)
DR	Lower left of aircraft symbol	Dead Reckoning–System is using projected position rather than GPS position to compute navigation data and sequence active flight plan waypoints



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Database management

Database

NOTE

Not all published Instrument Approach Procedures (IAP) are in the Navigation database. Pilots planning on flying an RNAV instrument approach must ensure that the Navigation database contains the planned RNAV Instrument Approach Procedure and that approach procedure must be loaded from the Navigation database into the FMS flight plan by its name.

The ARINC database

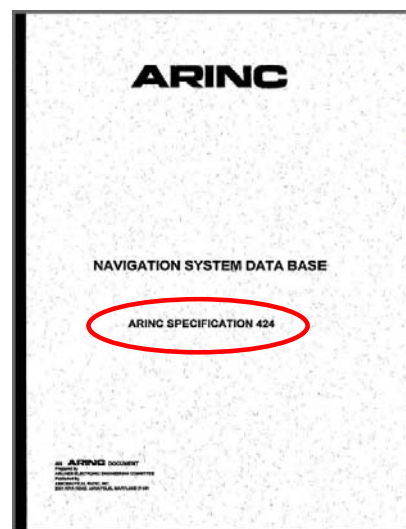
- ARINC = Aeronautical Radio, Inc.
- Principal stockholders: US scheduled airlines
- Other stockholders: other air transport companies, aircraft manufacturers, foreign flag carriers
- ARINC sponsors the Airline Electronic Engineering Committee (AEEC)
- AEEC formulates standards for electronic equipment and systems for airlines

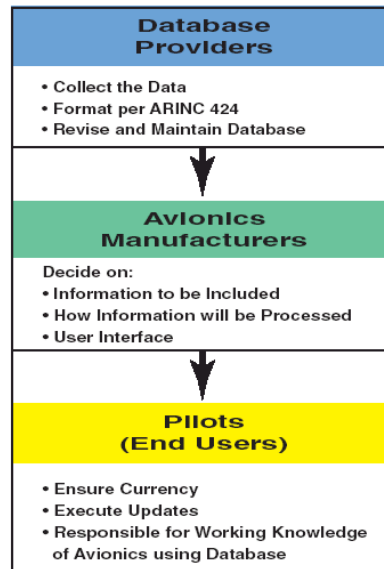
- Why a database?
- Reason 1:
 - To store permanent data, avoiding repetitive input of the same data

■ Reason 2:

- RNAV System is able to determine position relative to the earth's surface
 - but:
- it has to know the position of other points to be able to „navigate“
 - therefore:
- a „Directory of navigational fixes“ („Database“) is necessary

- “ARINC 424” specifies the format of navigation databases

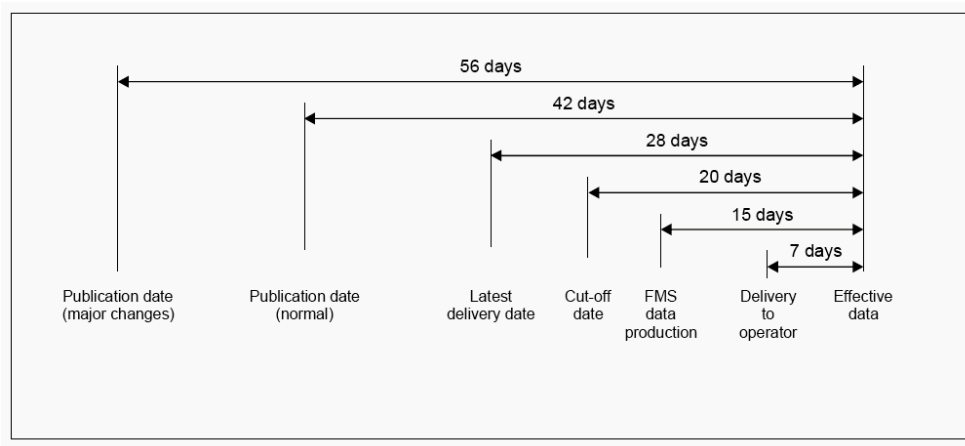




- The database must be valid for the current AIRAC cycle

- Database revision cycle
 - 28 Days (AIRAC Cycle) *)
 - Cutoff date:
 - Usually 20 days before the „effective date“
- *) **AIRAC: Aeronautical Information Regulation And Control**

- Data processing timeline



Operational assumptions

are made to accommodate aircraft performance:

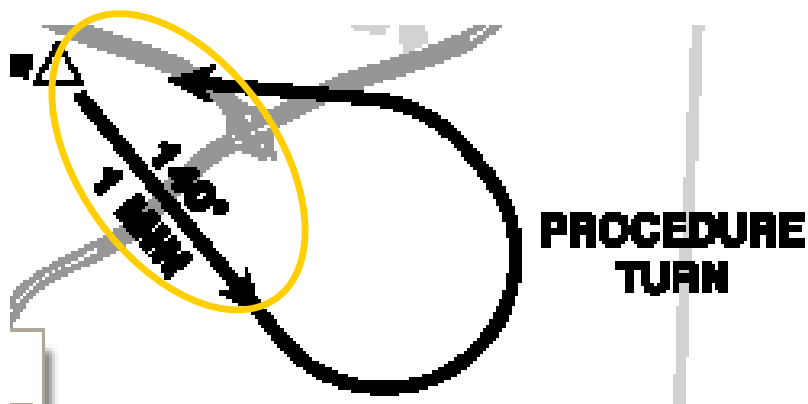
- speed - 210 Kt ground speed, 3.5 NM per min
- Max 25 degree bank angle used to compute turn radius
- Climb rate of 500 feet per NM
- Intercept angles - not specified, 30 degrees for intercept of localizer and 30 - 45 degrees for all others

- Stepdown fixes between the FAF and MAP are not included in navigation databases
- Lead radials are for non-RNAV equipped aircraft and are not intended to restrict the use of turn anticipation by the FMS

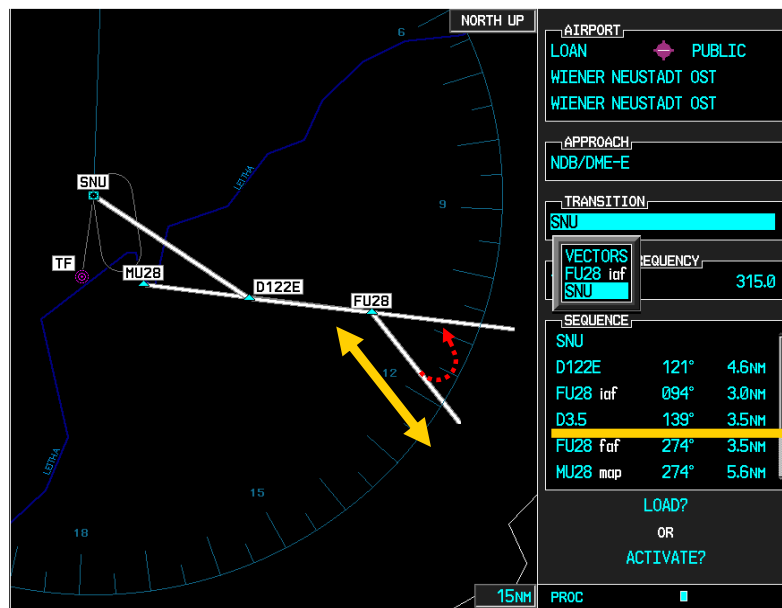
Most important for our operation:

- speed - 210 KT ground speed used to compute distance:
3.5 NM per min

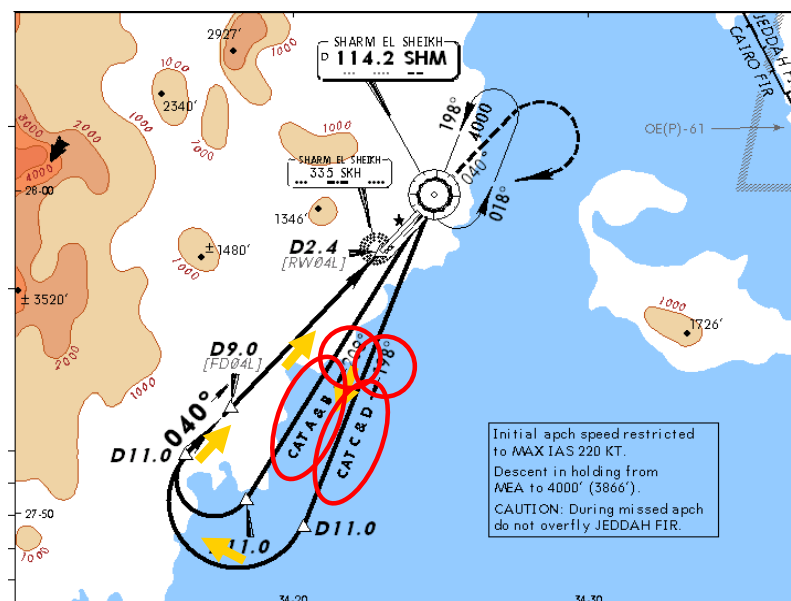
- „Old“ approach LOAN RWY 28



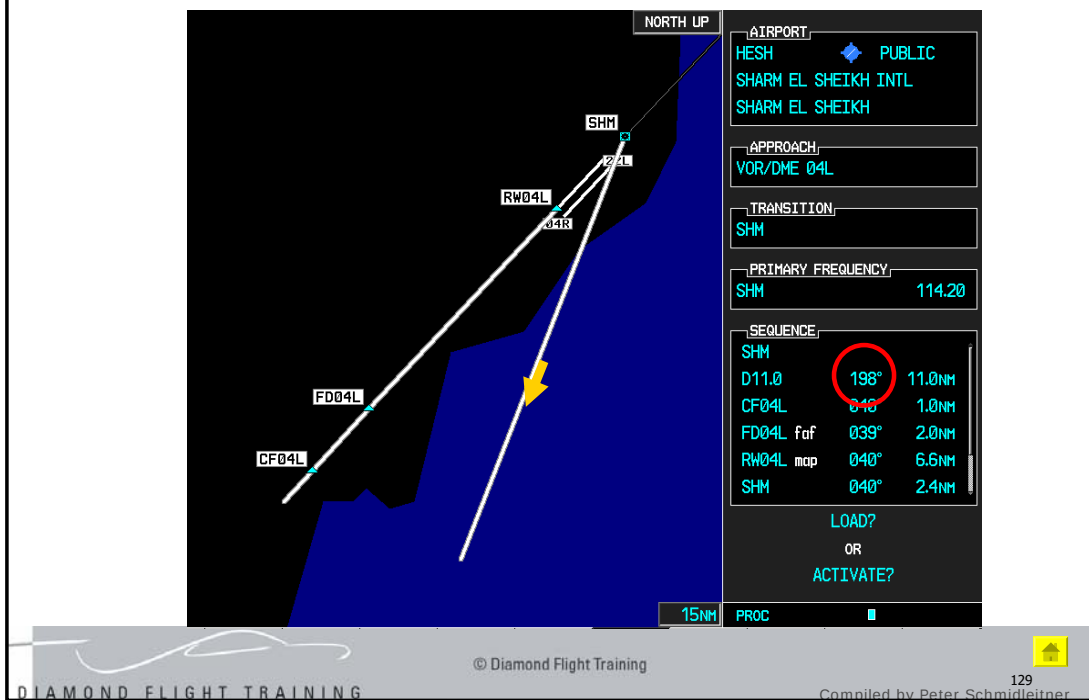
Database Operational assumptions



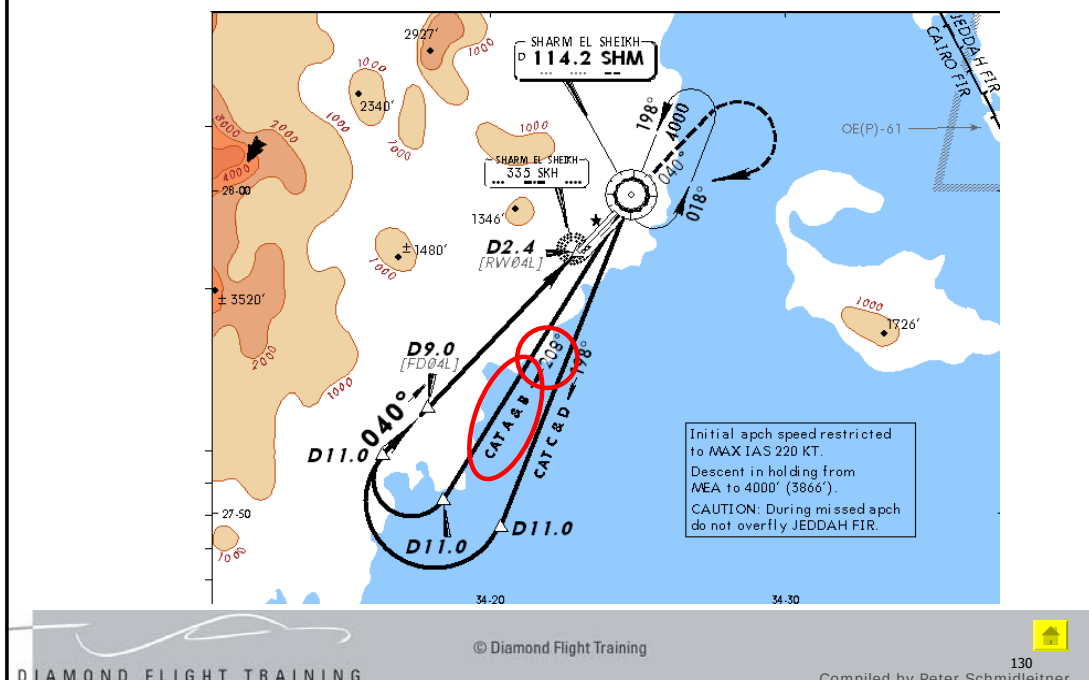
Database Operational assumptions



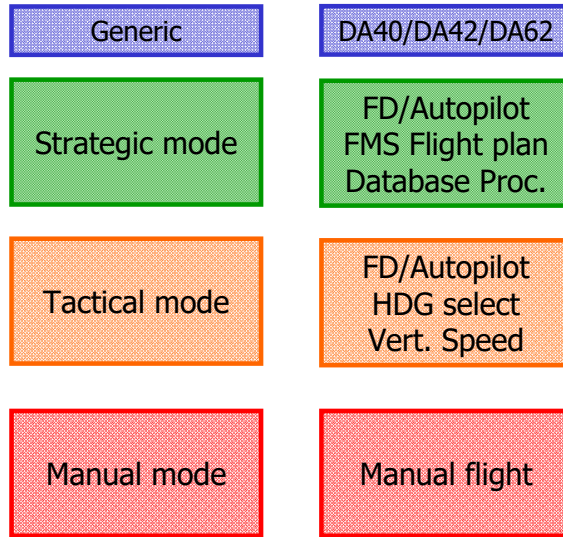
Database Operational assumptions



Database Operational assumptions



3 operational levels

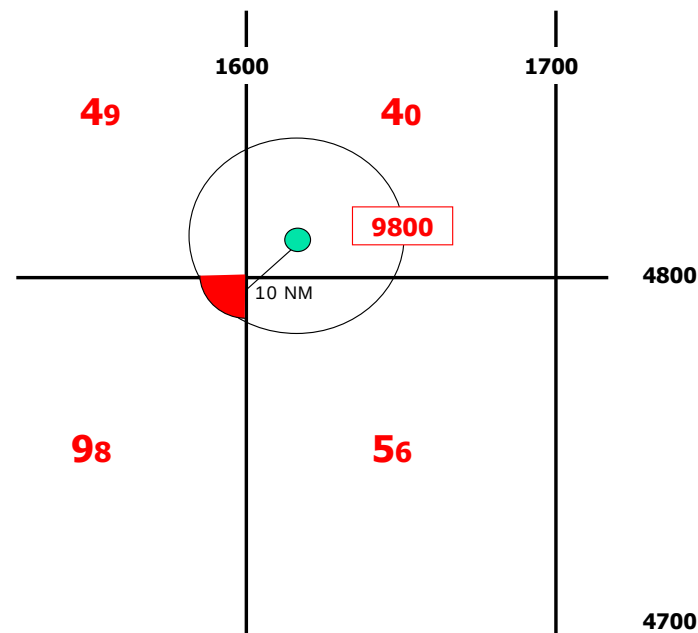


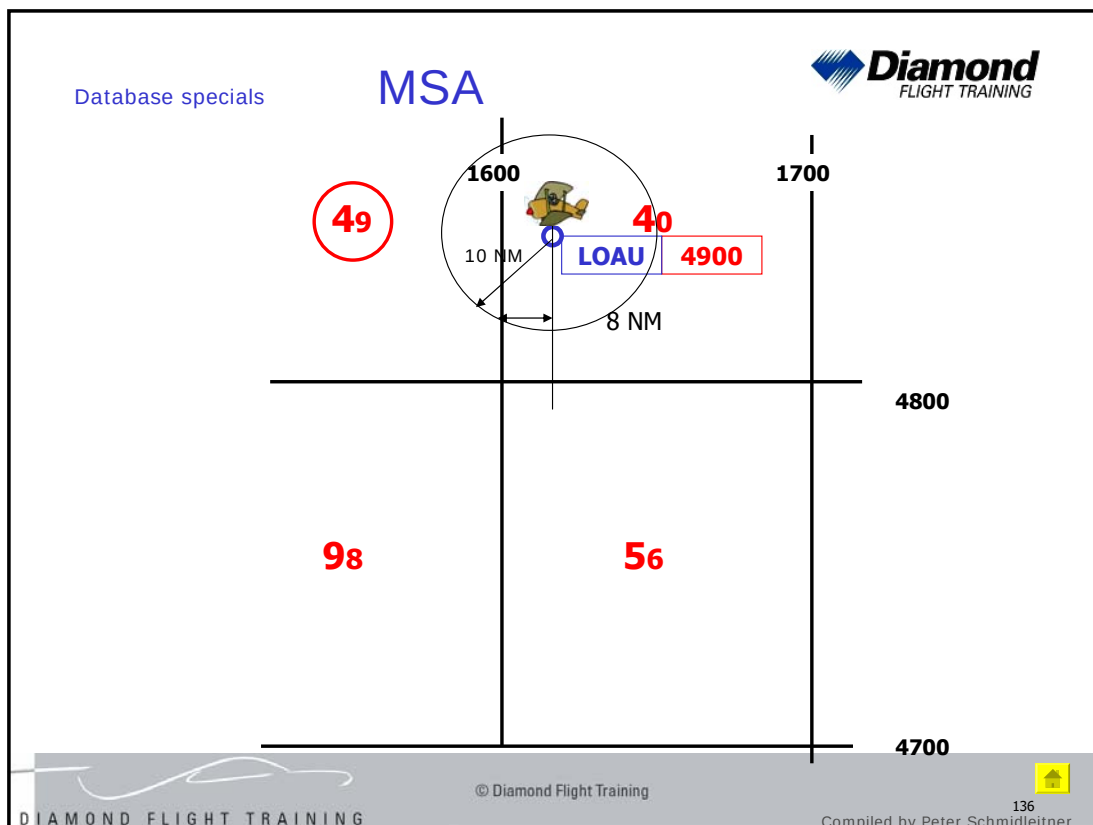
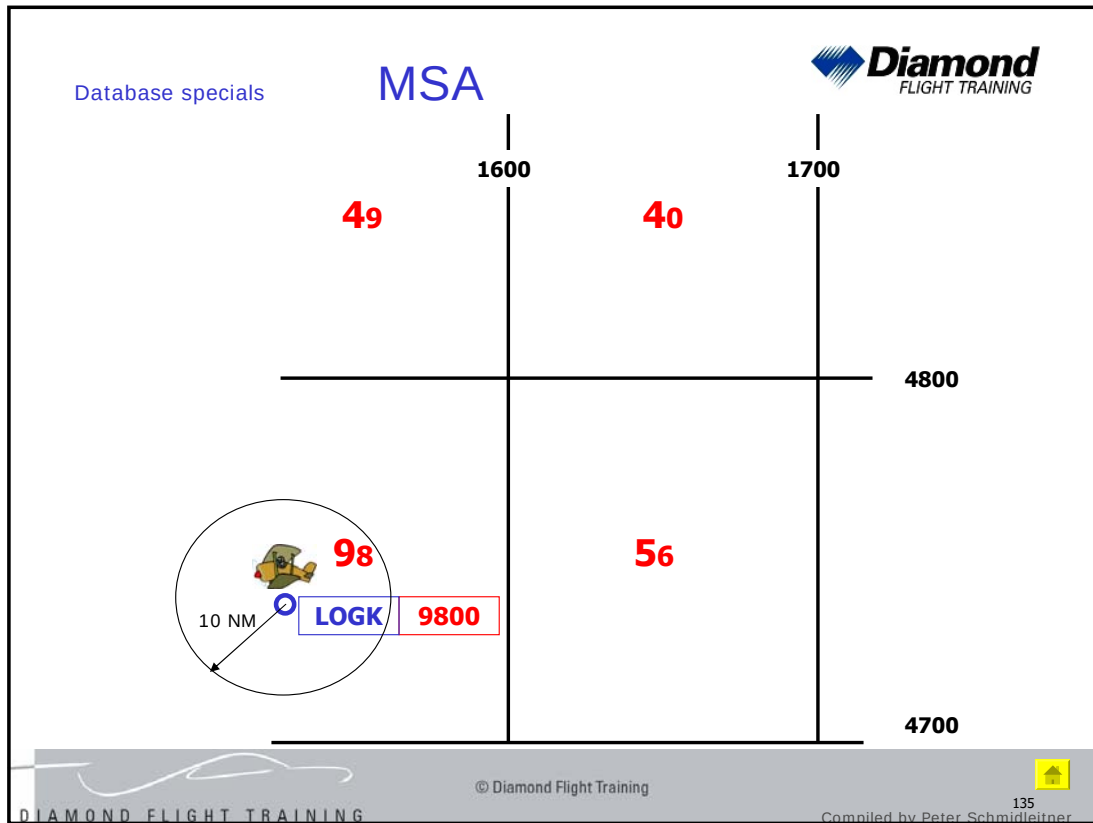
Use the right tool at the right time!

Some Database Specials

Minimum Safe Altitude

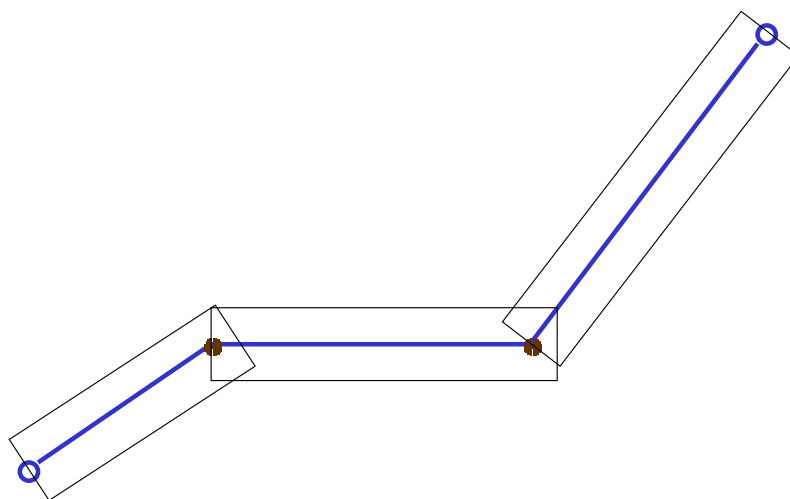
- Uses Grid Minimum Off-route Altitudes (Grid MORAs) to determine a safe altitude within a ten mile radius of your present position.
- The Grids are one degree latitude by one degree longitude in size.
- The highest elevation in the Grid, plus 1000ft or 2000ft (depending on the terrain) determines the Grid MORA.



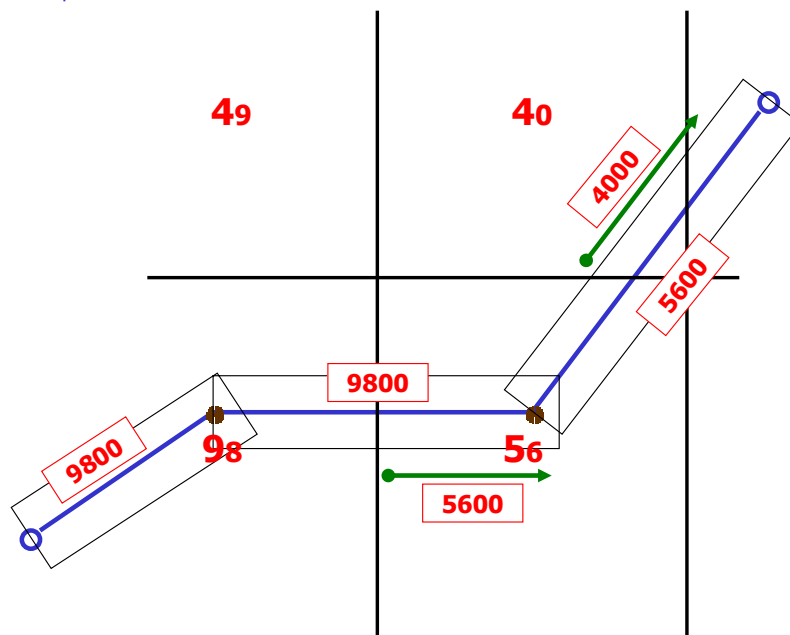
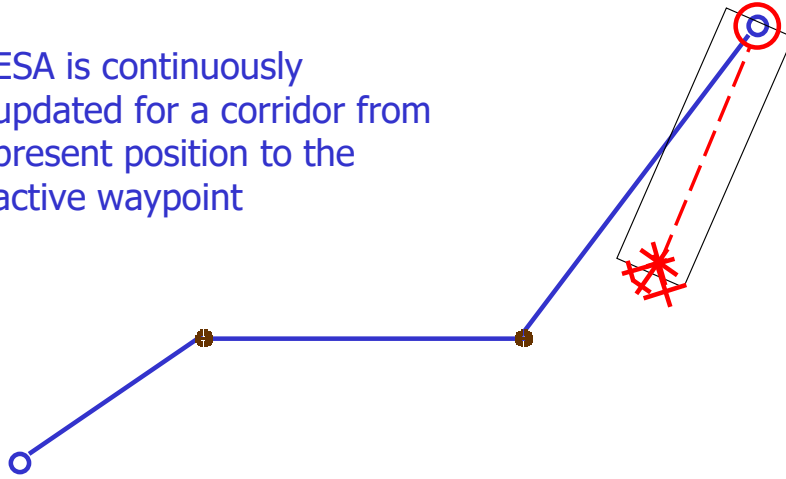


Enroute Safe Altitude

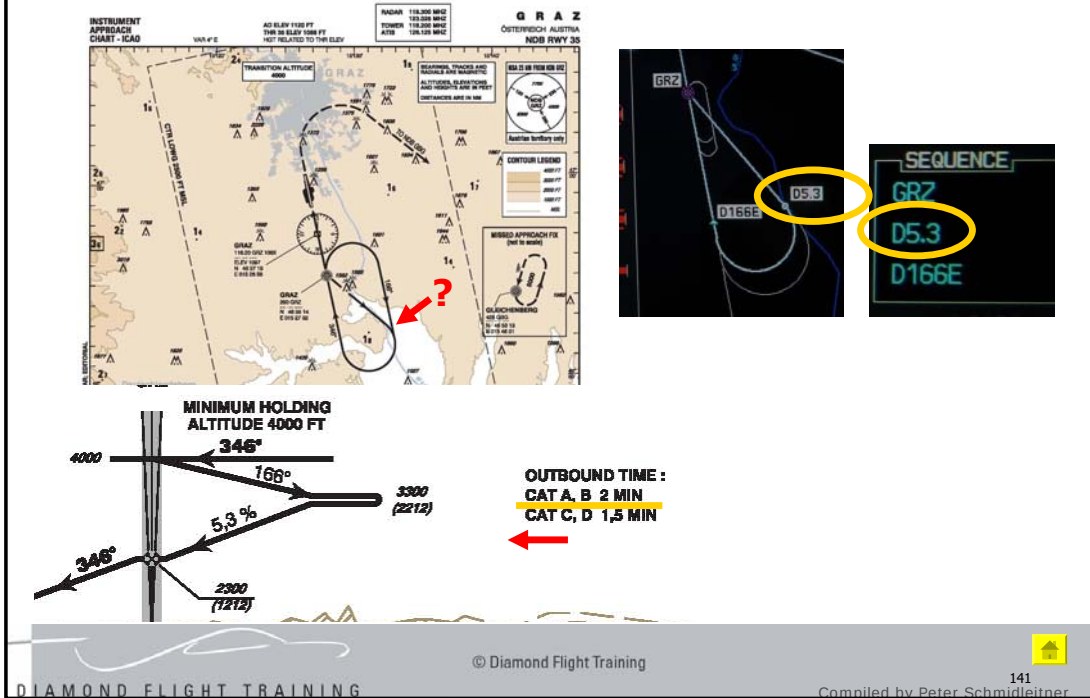
- The recommended minimum altitude within ten miles, left or right, of your desired course on an active flight plan or direct-to.
- Also uses the Grid MORA.



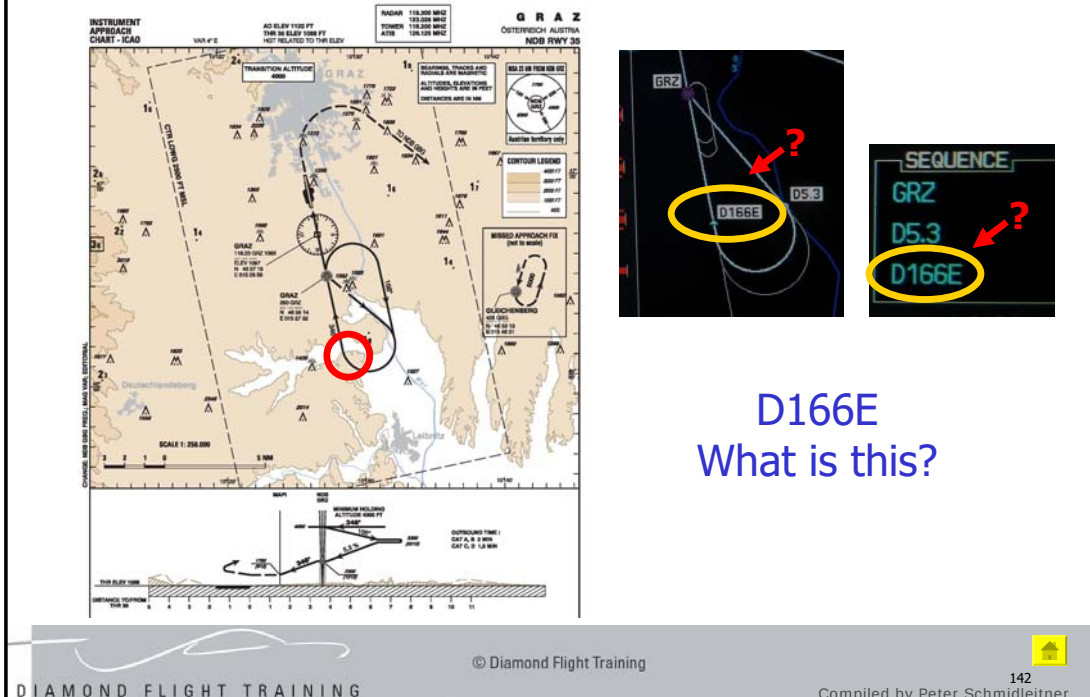
ESA is continuously updated for a corridor from present position to the active waypoint



Database specials „Unofficial“ waypoints



Database specials „Unofficial“ waypoints



D166E
What is this?

„Place“ – „Bearing“ – „Distance“

Bearing and Distance Waypoints

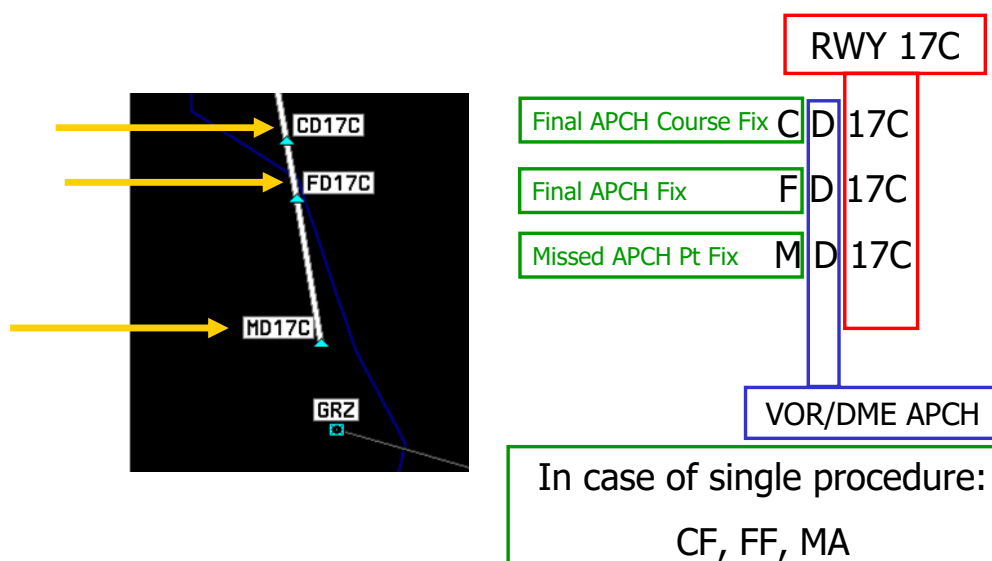
Identifiers should be developed by the application of the following rules:

1. The first character of the fix identifier should be „D“.
2. Characters two through four should signify the VHF **NAVAID radial** on which the waypoint lies.
3. The last character should be the DME arc radius defining the **position** of the waypoint radial. This radius should be expressed as the equivalent letter of the alphabet, i.e. A = 1nm, G = 7nm, O = 15NM, etc.

- „D“:
 - The waypoint is tied to some nearby navaid
- „Numbers xxx“
 - This is the direction from the navaid
- „Letters A, B, C, ...“
 - This is the distance from the navaid



A	B	C	D	E
1	2	3	4	5
F	G	H	I	J
6	7	8	9	10
K	L	M	N	O
11	12	13	14	15
P	Q	R	S	T
16	17	18	19	20
U	V	W	X	Y
21	22	23	24	25



Single Approach Procedure for a runway:

CF Final Approach Course Fix
FF Final Approach Fix
MA Missed Approach Point Fix

OM Outer Marker Fix

AF Initial Approach Fix
IF Intermediate Approach Fix
SD Step-Down Fix (or Sx if multiple)
RC Runway Centerline Fix
RW Runway Fix
MM Middle Marker Fix
IM Inner Marker Fix
BM Backcourse Marker Fix
TD Touchdown Fix inboard of runway threshold

Multiple Approach Procedures for a runway:

Fx, Ax, Ix, Cx, Mx, Sx, Rx, Tx where x is the "Route Type"

A	Approach Transition
B	LLZ Backcourse Approach
C	LORAN Approach
D	VOR/DME Approach
E	VOR Circle-To-Land Approach
F	FMS Approach
G	IGS (Instrument Guidance System) Approach
H	Helicopter Approach
I	ILS Approach
J	LLZ only Circle-To-Land Approach
K	LLZ Backcourse Circle-To-Land Approach
L	Localizer only Approach
M	MLS Approach
N	NDB Approach
P	GPS Approach
Q	NDB/DME Approach
R	RNAV Approach
S	VOR Approach with DME Facility
T	TACAN Approach
U	NDB Circle-To-Land Approach
V	VOR Approach (Non-DME Facility)
W	MLS Type A Approach
X	LDA (Localizer Directional Aid) Approach
Y	MLS Type B and C Approach
Z	SDF (Simplified Directional Facility) Approach

Requirements for RNAV and RNP specifications

Some of the Specification Requirements and special conditions

RNAV 10

- At least 2 independent and serviceable LRNSs, INS, IRS FMS or GNSS
- INS and IRU have time limitation, may be extended by updating

RNAV 5

- Manual data entry is acceptable

RNAV / RNP 1 and 2

- SID or STAR must be retrieved from on-board database
- Manual entry or creation of new waypoints (Lat/Long, PBD) not permitted

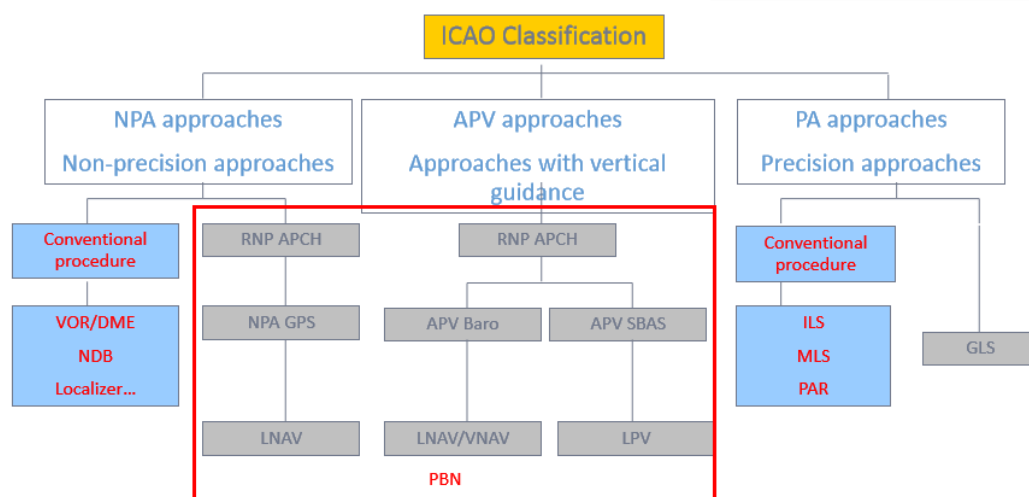
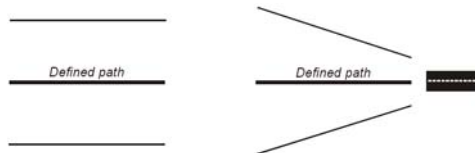
RNP 4

- At least 2 LRNSs (capable to RNP 4) must be operational when entering RNP airspace

RNP APCH

- Whilst for **oceanic, en-route and terminal phases** of flight, PBN is limited to operations with **linear** performance requirements, for the **approach phase**, PBN accommodates **both linear and angular** laterally and vertically guided operations.

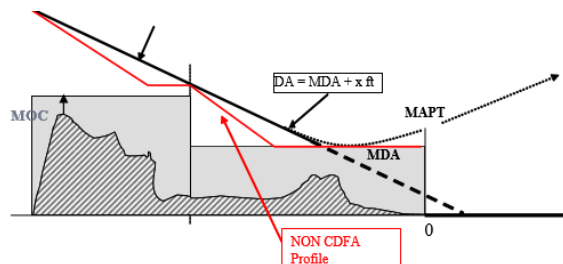
(However, vertical requirements do not constitute „Vertical PBN“, which is not defined or included in the PBN concept.)



- RNP APCH may be flown only when retrievable from the database
- RNP APCH to **LNAV minima** is a non-precision APCH, a **2D approach**
- RNP APCH to **LNAV/VNAV minima** has lateral guidance based on GNSS, vertical guidance based on SBAS or BARO/VNAV, it is a **3D approach**

- LNAV: Approach with lateral guidance
- APV: Approach with Vertical guidance:
(provides **CDFA**: Continuous Descent Final Approach)
 - LNAV/VNAV Approach
 - **LNAV+V Approach**
 - LPV Approach
(Localizer Performance with Vertical Guidance)
 - (PBN Manual classes these approaches as „RNP APCH operations“)

- **CDFA** does not need any particular on-board means
- Can be carried out using „vertical speed“ or „flight path angle“, or „Baro VNAV“



- **CDFA** no longer allows a level flight segment (at MDA) to the MAPt
- MDA cannot be used as a Decision Altitude (DA)
- Usually the „add-on concept“ is used:
 $DA = MDA + xx\text{Ft}$ (e.g. $xx = 50\text{Ft}$)

- An APV can be:
 - APV Baro
 - (LNAV/VNAV, LNAV+V)
 - APV SBAS
 - (LPV)

Note:

- Garmin (e.g. G1000) requires SBAS also for LNAV/VNAV, LNAV+V
 - “vertical guidance during an LNAV/VNAV or LNAV+V approach is based on GPS/SBAS altitude”

*ICAO **APV BARO** procedure design criteria now allow the use of SBAS for vertical guidance!*

This shall however be explicitly approved by the publishing ANSP before such an operation can be conducted.

The database coding must contain a special indicator to permit SBAS guidance for the APV BARO procedure.

Although depicted as a straight line on the approach chart, BARO-VNAV follows the earth's surface, implying a slightly curved glide path.

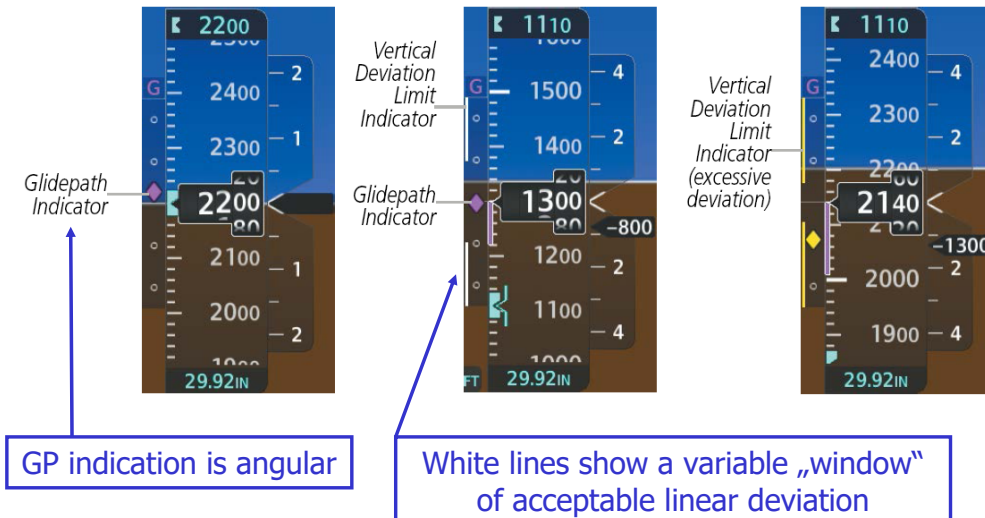
SBAS/GNSS VNAV on the other side calculates a straight glide path from the missed approach point (MAPt) to a point in space.

LNAV/VNAV approaches are constructed for APV BARO guidance; this assumes linear, not angular deviation indication.

How about the requirement to monitor that deviation above and below the vertical path must not exceed ± 75 feet?

How can this be applied with angular deviation indication?

The G1000 NXi solution:



G1000 Approach Types

HSI Annunciation	Description	Example on HSI
LNAV	GPS approach using published LNAV minima	 Approach Type - LNAV - LNAV+V - L/NAV - LPV
LNAV+V	GPS approach using published LNAV minima. Advisory vertical guidance is provided	
L/NAV	GPS approach using published LNAV/VNAV minima	
LPV	GPS approach using published LPV minima	

LNAV/VNAV

- An Approach may be an
 - **RNAV AR Approach**
 - AR: Authorisation Required (e.g. procedure contains RF-leg)
 - (in USA „SAAAR“: Special Aircraft and Aircrew Authorisation Required)

- **BARO VNAV operation:**
 - aerodrome temperature must be within promulgated range
- Correct altimeter setting is critical

- **RNP APCH to LPV minima:**

- A 3D operation
- Requires a FAS datablock

- The FAS data block contains the lateral and vertical parameters, which define the approach to be flown.
Each FAS data block ends with a CRC.
- CRC: Cyclic Redundancy Check
A mathematical algorithm applied to the digital expression of data that provides a level of assurance against loss or alteration of data.

Types of „Database Procedures“

- Genuine RNAV (e.g. GPS) procedures, using published waypoints
- „official“ overlay procedures (using published waypoints)
- „unofficial“ overlay procedures (conventional produres converted into „database language“)

Approaches based on GNSS are classified „RNP APCH“. They include existing RNAV(GNSS) approaches. Charting will retain the RNAV(GNSS) label for some time until standardisation can be achieved. Therefore the terms **RNP APCH** and **RNAV(GNSS)** will both exist.

Diamond
FLIGHT TRAINING



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173
Compiled by Peter Schmidleitner



**BARO-VNAV
OPERATIONS
NOT AUTHORIZED
BELOW -15°C**

OCA (OCH)	MAXIMUM GRADIENT	A	B	C	D
LNAV	2,5 %	1500 (420)			
LNAV/VNAV	2,5 %	1400 (320)			
LPV	2,5 %	1500 (420)	1500 (420)	1500 (420)	1500 (420)
	4 %	1350 (270)	1350 (270)	1350 (270)	1350 (270)

174

174
Compiled by Peter Schmidleitner

Database coding

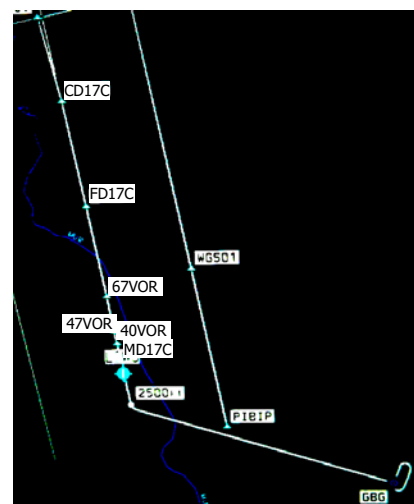
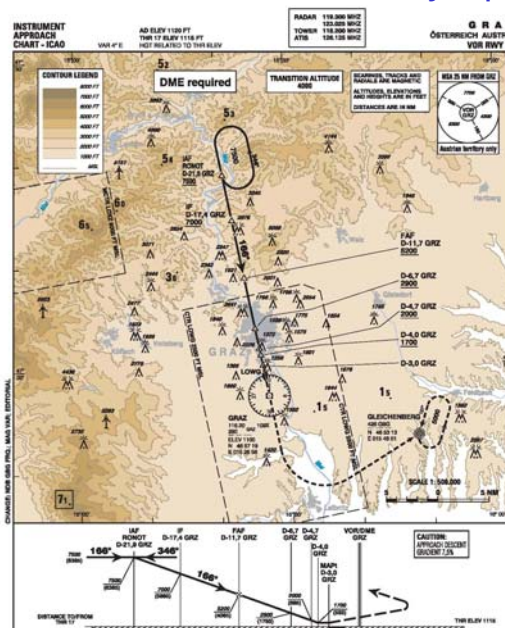
Instrument Approach Procedure Coding Table Graz RNAV (GNSS) RWY 35									
Path Transition	Identifier	Type	Frequency	Coordinates	Course of Track (° True)	DIST NM	Turn Direction	Level	Remarks
IF	WGR32	IAF	no	N464837.77 E0152256.29				A300+	RNP APCH
IF	WGR33	IAF	no	N464821.71 E0153022.98				A300+	RNP APCH
IF/TF	VAGL	IAF/IF	no	N464810.70 E0152511.98	056° (059.3°) 270° (279.3°)	5.0	left	A300+	RNP APCH
TF	VGR35	FAF	no	N465306.61 E0152607.39	348° (349.3°)	4.0		A300+	RNP APCH
TF	RW05	MAP	yes	N465040.03 E0152635.81	348° (349.3°)	5.7			RNP APCH
TF	VGR36	MAP	yes	N465036.46 E0152603.77	348° (349.3°)	2.0			RNP APCH
DF	GEG	MAP	yes	N465313.16 E0154801.15			right	A500+	RNP APCH

RNAV Holding							
Holding Point	Inbound Track True	Inbound Track MAG	Turn Direction	MAX IAS	Minimum Holding Altitude FT MSL/LC	Time	Remarks
GEG	200.3°	197°	left		A5000	1 MIN	

SBAS FAS Data Block Coding Table Graz RNAV (GNSS) RWY 35	
Input Data	
Parameters	Values
Operation Type	0
SBAS Provider	1
Airport Identifier	LOWG
Runway	35
Runway Direction	2
Approach Performance Designator	0
Route Indicator	
Reference Path Data Selector	0
Reference Path Identifier	035A
LTP/FTP Latitude	465840.0300N
LTP/FTP Longitude	0152635.8100E
FPAP Ellipsoidal Height (metres)	378.5
FPAP Latitude	470014.1460N
Delta FPAP Latitude (seconds)	94.1160
FPAP Longitude	0152609.9025E
Delta FPAP Longitude (seconds)	-25.9075
Threshold Crossing Height	53.0
TCH Units Selector	0
Glidepath Angle (degrees)	3.10
Course Width (metres)	107.00
Length Offset (metres)	224
HAL (metres)	40.0
VAL (metres)	50.0
Output Data	
Data Block	10 07 17 0F 0C A3 00 00 01 35 33 05 3C 22 29 14 44 A6 A0 06 C9 22 48 DF 02 99 35 9F 12 02 36 01 6C 1C 08 FA A6 38 SC AF
Calculated CRC Value	A6385CAF
Parameters	Values
ICAO Code	LO
LTP/FTP Orthometric Height (metres)	331.5
FPAP Orthometric Height (metres)	331.5

FAS Data Block

„Unofficial“ overlay approach



Cockpit Procedures

- Prior to commencing the approach (before the IAF), the pilot must verify the correctness of the loaded procedure by comparison with the appropriate approach charts.

This check must include:

- the waypoint sequence,
 - reasonableness of the tracks and distances of the approach legs, and the accuracy of the inbound course and mileage of the final approach segment, and
 - the vertical path angle.
- *Note: As a minimum, this check could be a simple inspection of a suitable map display.*

Intentionally blank

Phraseology

Aircraft	ATC
'(Aircraft c/s), request RNAV approach, via (IAF Designator), runway xx'	
	'(Aircraft c/s), cleared RNAV approach, runway xx, (report at [IAF designator])'
	'(Aircraft c/s), report established on final approach track'
	'(Aircraft c/s), report 2 miles from final approach fix'
	'(Aircraft c/s), report final approach fix'

Aircraft	ATC
	'(Aircraft c/s), GNSS reported unreliable'
	'(Aircraft c/s), GNSS may not be available due to interference in the vicinity of (location) (radius) [between (levels)] OR '... In the area of (description) [between (levels)]'
	'(Aircraft c/s), GNSS unavailable for (specify operation) [from (time) to (time) (or until further notice)]'
'(Aircraft c/s) Unable RNP (specify type) (or RNAV) (due to [reason eg Loss of RAIM or RAIM alert]) (plus intentions)'	
'(Aircraft c/s) Loss of RAIM or RAIM alert (plus intentions)'	

Flight Management System (FMS)

FMS

Flight Management System (FMS)

- An integrated system,
- consisting of
 - airborne sensor,
 - receiver and
 - computer
- with both
 - navigation and
 - aircraft performance databases,
- which provides
 - performance and
 - area navigation guidance
- to a display and the automatic flight control system.

- A primary function is in-flight management of the flight plan.
- FMS may also contain company (user) flight plan routes and user waypoints.
- Flight Management Systems in transport category aircraft are multi sensor systems.
- From the cockpit, the FMS is normally controlled through a Control Display Unit (CDU) which incorporates a small screen and keyboard or touchscreen.

A sample CDU:

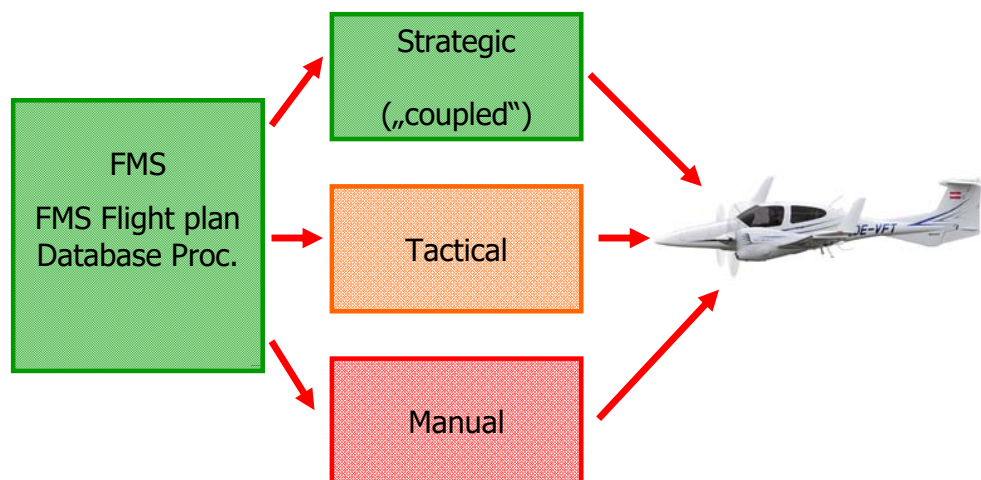


- The FMS does not „steer“ the aircraft
- The aircraft must accept the FMS commands
 - by manual,
 - semi automatic, or
 - fully Automatic
- inputs

Do you remember the „3 Operational Levels“?:

Strategic, Tactical, Manual

Never forget the FMS „Connection“ to the ACFT



ICAO Flight Plan

ICAO Flight Plan

AFM Supplement A33:

- the P-RNAV Item 10a code does no longer exist;
- the meaning of the Item 10a code „R” is “PBN approved”;
- specific PBN capabilities are amplified in Item 18;
- flight plans may be rejected if R is filed in Item 10a and no PBN information is filed in Item 18.
- in item 10 code “B” indicates the LPV capability
- for DA62 the correct Item 18 entry would be:

A1L1B2C2D2S1S2

Specification	Reference Document		ICAO Flight Plan Code	Integrated flight deck G1000 with SBAS
	FAA	EASA or JAA		
RNAV 10 (RNP 10) Oceanic	FAA Order 8400.12B	-	A1	Yes
B-RNAV/RNAV 5 (operations in Europe)	FAA AC 90-96A CHG 1	EASA AMC 20-4	B2	Yes
RNAV 2	FAA AC 90-100A	-	C2	Yes
RNAV 1	FAA AC 90-100A	-	D2	Yes
P-RNAV (operations in Europe)	FAA AC 90-96A CHG 1	JAA TGL 10 Rev 1	D2	Yes
RNP 4 (Oceanic)	FAA Order 8400.33	-	L1	Yes
RNP 1	FAA AC 90-105	-	O2	No
RNP APCH LNAV	FAA AC 90-105	EASA AMC 20-27	S1	Yes
RNP APCH LNAV/VNAV	FAA AC 90-105	EASA AMC 20-27 with CM-AS-002	S2	Yes
LP	FAA AC 90-107	-	N/A	Yes (Note 1)
LPV	FAA AC 90-107	EASA AMC 20-28	N/A	Yes
RNP AR APCH	FAA AC 90-101A	EASA 20-26	T1	No

Note 1: When GDU software version 13.00 or later is installed.